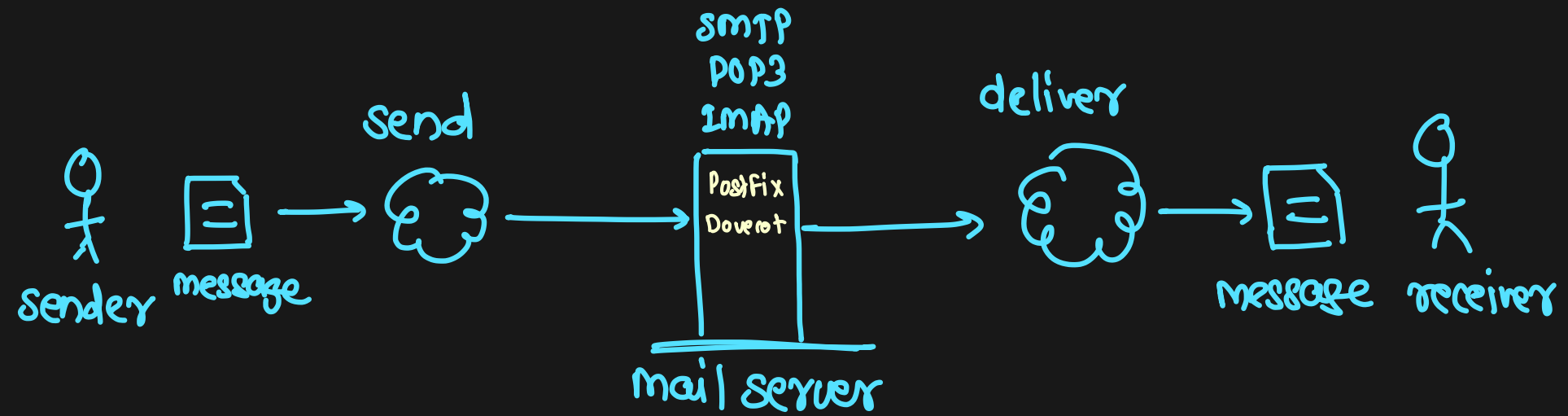




e Mail Servers

Mail Services



Postfix

Dovecot

Maildir

- A mail service is a system that enables sending, receiving, storing, and managing email messages over a network
- They typically follow a client-server model using standard protocols
- It works using a combination of:
 - Protocols : SMTP | IMAP | POP3
 - Servers : Postfix | Dovecot | Procmail | Sendmail
 - Clients : mail | mailx | thunderbird | outlook | Apple mail

Envelope Information → used to send the message



Part Field	Example	Purpose
<u>From</u>	Alice <alice@gmail.com>	<u>Sender's name and email</u>
<u>To</u>	Bob <bob@company.com>	<u>Primary recipient(s)</u>
<u>Cc</u>	Carol@company.com	<u>Carbon copy recipients</u> → visible
<u>Bcc</u>	(hidden from recipients)	<u>Blind carbon copy</u> → invisible
<u>Subject</u>	"Meeting tomorrow"	<u>Brief summary of content</u>
<u>Date</u>	Mon, Apr 13, 2026 10:30 AM	<u>When it was sent</u>
<u>Attachments</u>	📎 meeting_notes.pdf	<u>Files attached to the email</u>

Body

actual message

Raw Email Format - RFC 5322

[Headers]

From: alice@gmail.com

To: bob@company.com

Subject: Meeting tomorrow

Date: Mon, 13 Apr 2026 10:30:00 -0700

Message-ID: <12345@mail.gmail.com>

[Blank line]

[Body]

Hello Bob,

Let's meet at 2pm.

Thanks,
Alice

Email

→ headers

→ From

→ to

→ cc / bcc

→ body → message

A hand-drawn diagram of an email client interface. It features a rectangular box containing four input fields labeled 'From:', 'to:', 'Subject:', and 'body'. Each label is followed by a horizontal line representing an input field. Below the 'body' field is a button labeled 'send'.

Email client

Components of Email

■ Mail user agent (MUA) → mail client

- The component of the e-mail system that the user sees and interacts with, such as the Thunderbird, Outlook, Evolution, or Mutt program
- An MUA is responsible only for reading mail and allowing users to compose mail
- Examples: mail/mailx, mutt, gmail UI

■ Mail transport agent (MTA) → mail relay

- Handles the process of getting the mail from one site to another
- The transmission details are specified by SMTP
- Also known as message transfer agent and mail relay
- E.g. Postfix, Exim, Sendmail

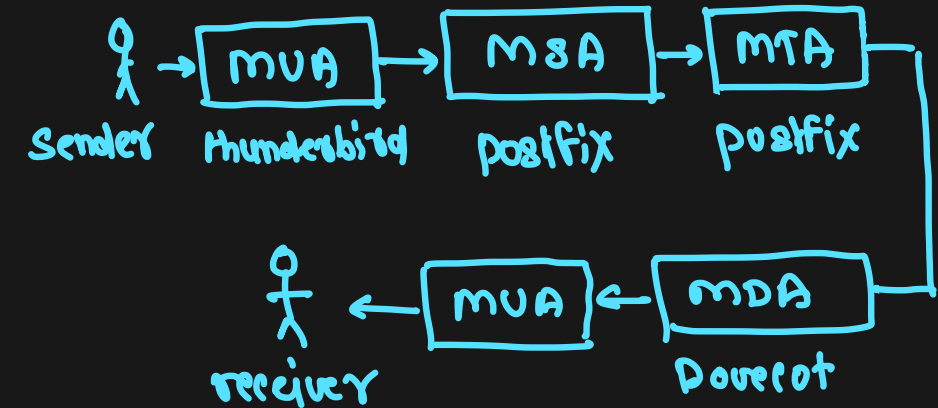
■ Mail delivery agent (MDA)

- Responsible for distributing and sorting any received messages on the local machine to the appropriate user mailbox
- Also known as local delivery agent (LDA)
- Examples: Dovecot, Procmail, Cyrus IMAP, fetchmail, getmail, fdm etc.

■ Mail Submission Agent (MSA)

- Accepts mail from MUAs (often part of MTA)
- Example: Postfix on port 587

→ online → mail.google.com
→ offline → Thunderbird, apple mail



Mail storage formats



- Mail storage formats define how email messages are stored on disk by the Mail Delivery Agent (MDA) and accessed by the Mail User Agent (MUA) or IMAP/POP3 server
- The choice affects performance, reliability, and compatibility

Feature	<u>mbox</u> ✗	<u>Maildir</u> ✓
<u>Messages per file</u>	<u>All in one file</u>	<u>One file per message</u>
<u>Locking required</u>	<u>Yes (flock, fcntl, dot-lock)</u>	<u>No (filename-based)</u>
<u>NFS safety</u>	<u>Poor</u>	<u>Good</u>
<u>Backup granularity</u>	<u>Whole mailbox</u>	<u>Per message</u>
<u>Deletion speed</u>	<u>Slow (rewrites entire file)</u>	<u>Fast (unlinks single file)</u>
<u>Corruption risk</u>	<u>High (single file corruption)</u>	<u>Low (only one message lost)</u>

Maildir format



- Standard Maildir (qmail-style)
- Structure:
 - A directory with three subdirectories:
 - ~/Maildir/
 - | | | |
|--|-------------|--|
| | <u>cur/</u> | # <u>Messages already seen/processed</u> |
| | <u>new/</u> | # <u>New messages not yet seen</u> → <u>unseen</u> |
| | <u>tmp/</u> | # <u>Temporary files during delivery</u> |
- Filename format:
 - [unique] [info] , [host] : [flags]
 - E.g. 1234567890.12345_0.hostname,S=1024,W=2048:2,S
 - Flags
 - D: Draft
 - F: Flagged - Starred/Important
 - P: Passed/Forwarded
 - R: Replied
 - S: Seen (Read)
 - T: Trashed - In Trash

Working of Email



■ Step 1: User Composes and Sends

- Alice opens Gmail (MUA) and writes an email to `bob@company.com`
- She clicks Send

■ Step 2: MUA to MSA (Submission)

- Alice's MUA connects to Gmail's MSA on port 587 (SMTP with STARTTLS)
- It authenticates using Alice's credentials
- The MSA performs initial checks:
 - Is Alice authorized to send?
 - Is the message size within limits?
 - Does it contain obvious spam?

■ Step 3: MSA to MTA (Relay)

- Gmail's MSA passes the message to Gmail's MTA
- The MTA adds headers:
 - Received: from ... (trace headers)
 - Date: ...
 - Message-ID: <unique@mail.gmail.com>

Working of Email



■ Step 4: DNS Lookup (Finding the Destination)

- Gmail's MTA extracts the domain: company.com
- It performs a DNS MX (Mail Exchange) query
- MX records show priority (lower = higher priority): 10 mail.company.com
- Then resolves mail.company.com to an IP address (A record)

■ Step 5: SMTP Conversation (Server to Server)

- Gmail's MTA connects to mail.company.com on port 25 (SMTP)
- They perform an SMTP conversation

■ Step 6: MTA to MDA (Delivery)

- Company's MTA (Postfix) receives the message
- It hands the message to the MDA (e.g., Dovecot, Procmal)
- The MDA:
 - Applies filters (spam, antivirus)
 - Checks recipient exists (bob)
 - Writes the message to Bob's mailbox in mbox or Maildir format

■ Step 7: Storage

- The message is stored on disk:
 - mbox format: /var/mail/bob (single file)
 - Maildir format: /home/bob/Maildir/new/ (one file per message)
- Step 8: Retrieval by Recipient
- Bob opens Outlook (MUA) on his computer
- Outlook connects to Company's IMAP server (Dovecot) on port 993 (IMAPS)
- Bob authenticates
- The server sends a list of messages in Bob's INBOX
- Bob reads the message (server marks it as \Seen)

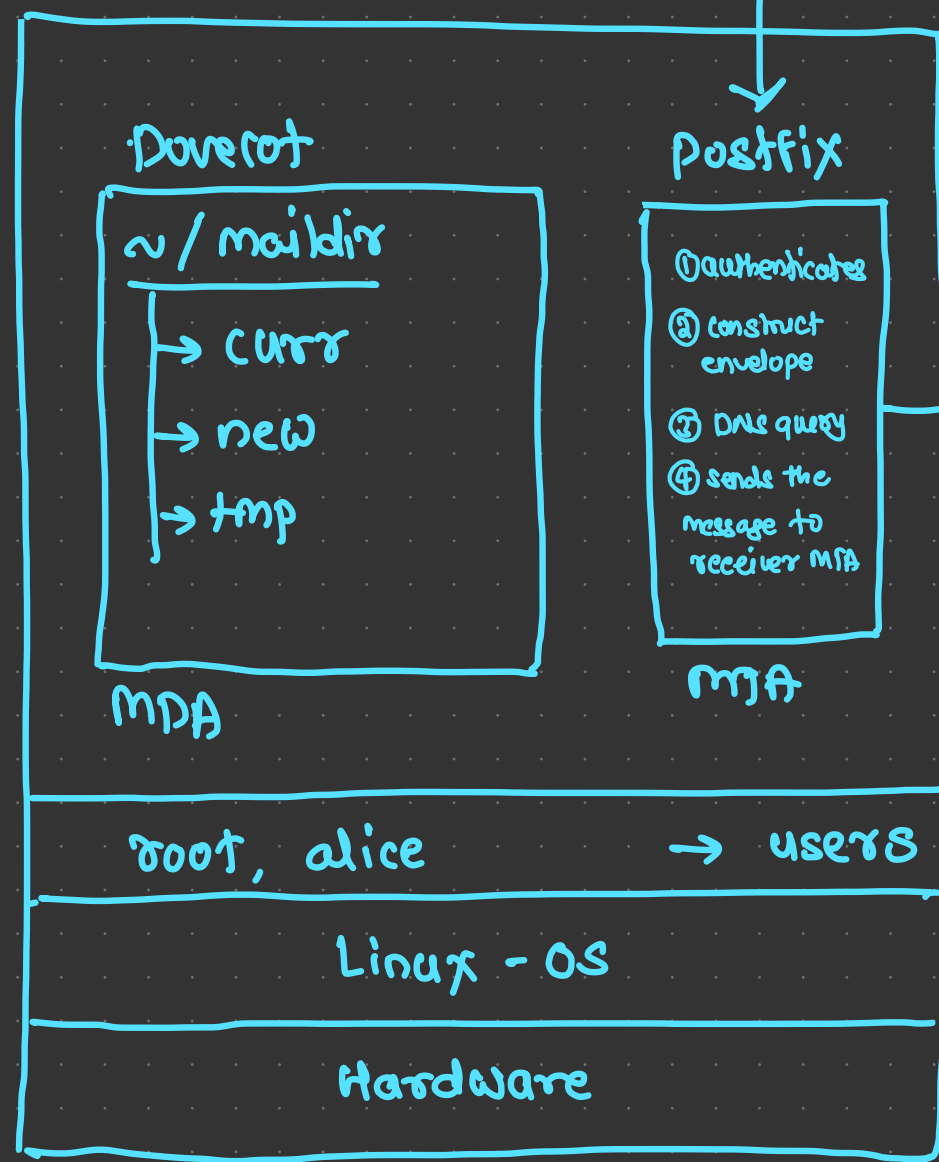
alice

alice@gmail.com

receiver: -
subject: -
body: -

MUA → gmail.com

port 587



mail.google.com

18.7.8.9

DNS server

mx yahoo.com mail.yahoo.com
A mail.yahoo.com 20.5.6.7

- ① find mail server of yahoo.com (mx) → mail.yahoo.com
- ② what is IP address of mail.yahoo.com (A) → 20.5.6.7

- ③ Gmail MIA communicates & sends email to yahoo MIA

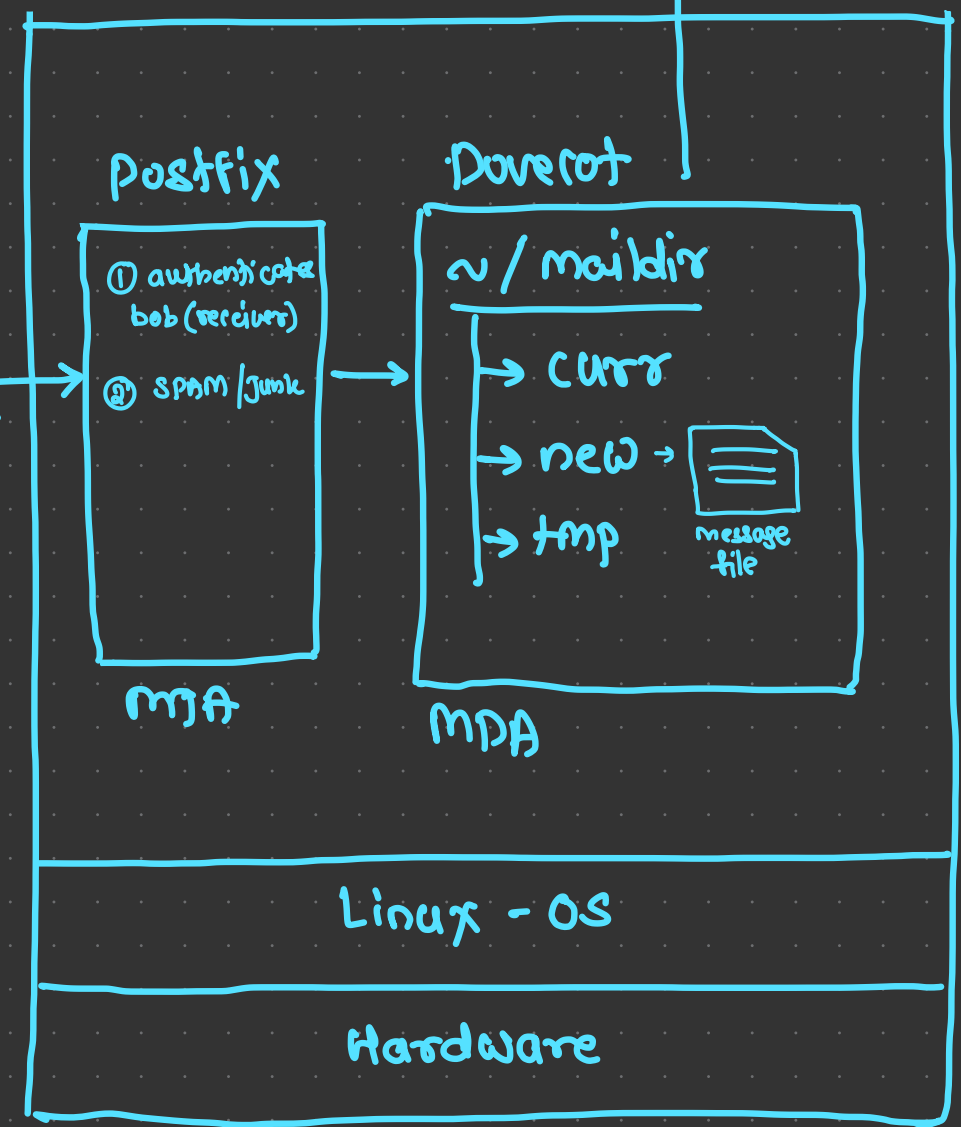
port 25

bob@yahoo.com

bob

message1
message2
message3

MUA (mail.yahoo.com)



mail.yahoo.com

20.5.6.7

Common Scenarios



- **Scenario A: Both users on same server**
 - Alice@company.com → Bob@company.com
 - MTA recognizes local domain
 - Bypasses DNS/network hop
 - Directly hands to MDA
- **Scenario B: User offline (no IMAP/POP3 access)**
 - Server stores message in mailbox
 - When Bob finally connects (hours/days later)
 - IMAP server delivers stored messages
- **Scenario C: Greylisting (anti-spam)**
 - Receiving server returns temporary failure (450)
 - "I don't know you, try again later"
 - Legitimate servers retry (spammers often don't)
 - On retry (minutes later), server accepts
- **Scenario D: Bounce handling**
 - Invalid recipient (bob@company.com doesn't exist)
 - Receiving MTA returns 550 No such user
 - Sending MTA generates DSN (Delivery Status Notification)
 - Alice receives bounce message



SMTP

postfix

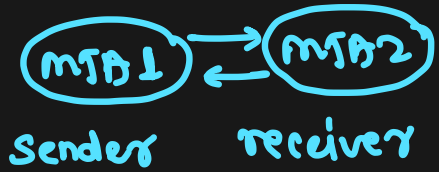
SMTP



- SMTP's origins began in 1980, building on concepts implemented on the ARPANET since 1971
- The Simple Mail Transfer Protocol (SMTP) is the de facto standard for transporting mail around the Internet
- Anyone who wants to have a mail server capable of sending and receiving mail must be able to support SMTP
- As a standards-based protocol (see RFC 5321), SMTP is well understood, platform independent, and well supported across a variety of operating systems and devices

Port	Name	Encryption	Typical Use
✓ <u>25</u>	<u>SMTP</u>	<u>None (STARTTLS optional)</u>	<u>MTA to MTA relaying</u>
✓ <u>587</u>	<u>Submission</u>	<u>STARTTLS (mandatory recommended)</u>	<u>MUA to MTA (authenticated)</u>
✓ <u>465</u>	<u>SMTPS</u>	<u>Implicit TLS (deprecated but used)</u>	<u>Legacy MUA submission</u>

SMTP Commands



Command	Syntax	Purpose
① <u>HELO</u>	<u>HELO hostname</u>	<u>Initiate session (basic)</u>
② <u>EHLO</u>	<u>EHLO hostname</u>	<u>Initiate session with extensions</u>
③ <u>MAIL FROM</u>	<u>MAIL FROM:<sender></u>	<u>Specify sender</u>
④ <u>RCPT TO</u>	<u>RCPT TO:<recipient></u>	<u>Specify recipient (multiple allowed)</u>
⑤ <u>DATA</u>	<u>DATA</u>	<u>Begin message content</u>
<u>RSET</u>	<u>RSET</u>	<u>Abort current transaction</u>
<u>VERFY</u>	<u>VERFY <user></u>	<u>Verify user exists (often disabled)</u>
<u>EXPN</u>	<u>EXPN <list></u>	<u>Expand mailing list (often disabled)</u>
<u>NOOP</u>	<u>NOOP</u>	<u>Do nothing (keep-alive)</u>
⑥ <u>QUIT</u>	<u>QUIT</u>	<u>End session</u>

SMTP Response

Code	Meaning	Example
<u>2xx</u>	<u>Success</u>	<u>250 OK</u>
<u>3xx</u>	<u>Additional info needed</u>	<u>354 Start mail input</u>
<u>4xx</u>	<u>Temporary failure (retry later)</u>	<u>450 Mailbox busy</u>
<u>5xx</u>	<u>Permanent failure</u>	<u>550 No such user</u>



```
$ telnet mail.example.com 25
220 mail.example.com ESMTP Postfix
C EHL0 client.example.com
R 250-mail.example.com
R 250-PIPELINING
R 250-SIZE 10240000
R 250-STARTTLS
R 250-AUTH PLAIN LOGIN
R 250-8BITMIME
R 250 DSN
C MAIL FROM:<alice@example.com>
R 250 2.1.0 Ok
C RCPT TO:<bob@remote.com>
R 250 2.1.5 Ok
C DATA
R 354 End data with <CR><LF>.<CR><LF>
  Subject: Test
  .
R 250 2.0.0 Ok: queued as 12345
QUIT
221 2.0.0 Bye
```

SMTP Working



- SMTP defines the method by which mail is sent from one host to another
- It does not define how the mail should be stored. It does not define how the mail should be displayed to the recipient
- SMTP was the first mail standard that was independent of the transport mechanism. This meant people using TCP/IP networks could use the same format to send a message as someone using two cans and a string—at least theoretically
- SMTP is also independent of operating systems, which means each system can use its own style of storing mail without worrying about how the sender of a message stores her mail
- In the Free Open Source Software (FOSS) world, several software packages (such as Exim, Postfix, Sendmail, and opensmtpd) provide their own implementation of SMTP



- Install postfix

- sudo yum install postfix

- Configuration

- sudo vim /etc/postfix/main.cf

myhostname = <hostname>.example.com

mydomain = example.com

myorigin = \$mydomain

inet_interfaces = all

mydestination = \$myhostname, localhost.\$mydomain,
localhost, \$mydomain

mynetworks = 192.168.1.0/24, 192.168.96.0/24, 127.0.0.0/8

home_mailbox = Maildir/

- Start server

- sudo systemctl enable postfix
 - sudo systemctl start postfix

- Firewall Configuration

- sudo firewall-cmd --add-service=smtp --permanent
 - sudo firewall-cmd --reload



IMAP/POP3

Dovecot

POP3



- **RFC:** 1939
- **Role:** Download emails from server to client (typically deletes from server)
- **Characteristics:** Simple, stateful, limited server-side management

Port	Encryption	Use
<u>110</u>	<u>Plain (STARTTLS optional)</u>	<u>Standard POP3</u>
<u>995</u>	<u>Implicit TLS (POP3S)</u>	<u>Secure POP3</u>

▪ **Limitations**

- No folders (only one INBOX)
- No message flags (read/unread lost after download)
- No partial fetch (must download entire message) → once message gets downloaded, it deleted from server
- No server-side searching
- State management primitive

POP3 Commands

	Command	Syntax	Purpose
①	<u>USER</u>	<u>USER username</u>	<u>Specify username</u>
②	<u>PASS</u>	<u>PASS password</u>	<u>Send password</u>
	<u>APOP</u>	APOP name digest	Authenticate with MD5 (deprecated)
③	<u>STAT</u>	STAT	Mailbox status (count + size)
④	<u>LIST</u>	LIST [msg]	List message(s) and sizes
⑤	<u>RETR</u>	<u>RETR msg</u>	<u>Retrieve message</u>
	<u>DELE</u>	<u>DELE msg</u>	<u>Mark message for deletion</u>
	<u>RSET</u>	RSET	<u>Unmark all deletions</u>
	<u>TOP</u>	TOP msg n	Get headers + first n lines
	<u>UIDL</u>	UIDL [msg]	Get unique identifier list
⑥	<u>QUIT</u>	QUIT	End session (commits deletions)

```
$ telnet mail.example.com 110
+OK Dovecot POP3 ready

C USER alice
R +OK
C PASS secret
R +OK Logged in
C STAT
R +OK 2 5432
C LIST
R +OK 2 messages
  { 1 3200
    2 2232
  }
  .
C RETR 1
R +OK 3200 octets
  { Return-Path: <bob@example.com>
    From: Bob <bob@example.com>
    Subject: Hello
  }
  Message body...
  .
C DELE 1
R +OK Marked for deletion
C QUIT
R +OK Logging out, messages deleted
```



IMAP (Internet Message Access Protocol)

- **RFC:** 3501 (IMAP4rev1), 9051 (current)
- **Role:** Remote mailbox management – messages stay on server
- **Characteristics:** Complex, stateful, rich feature set

Port	Encryption	Use
<u>143</u>	<u>Plain (STARTTLS optional)</u>	<u>Standard IMAP</u>
<u>993</u>	<u>Implicit TLS (IMAPS)</u>	<u>Secure IMAP</u>

POP3 vs IMAP



Feature	SMTP	POP3	IMAP
<u>Direction</u>	<u>Push</u>	<u>Pull (download)</u>	<u>Pull (sync)</u>
<u>Message storage</u>	<u>Temporary (queues)</u>	<u>Local client only</u>	<u>Server + optionally local</u>
<u>Mailbox access</u>	N/A	<u>Single INBOX</u>	<u>Multiple folders</u>
<u>Message flags</u>	N/A	<u>No</u>	<u>Yes (server-side)</u>
<u>Partial fetch</u>	N/A	<u>No</u>	<u>Yes</u>
<u>Server search</u>	N/A	<u>No</u>	<u>Yes</u>
<u>Concurrent access</u>	N/A	<u>No (single client)</u>	<u>Yes</u>
<u>Offline support</u>	N/A	<u>Yes</u>	<u>Limited (requires sync)</u>
<u>Bandwidth</u>	<u>Moderate</u>	<u>High (full downloads)</u>	<u>Low (headers only possible)</u>
<u>Stateful</u>	<u>No</u>	<u>Yes (per session)</u>	<u>Yes (per session)</u>

Protocol Selection Guide



- **Use SMTP (only)**
 - Sending notifications from applications
 - Internal mail relay between servers
- **Use POP3 when**
 - Single device only
 - Limited server storage
 - Always-online not required
 - Simple requirements
- **Use IMAP when** → preferred
 - Multiple devices (phone, laptop, webmail)
 - Need server-side folders and flags
 - Collaboration (shared mailboxes)
 - Low bandwidth optimization needed
- **Modern standard stack**
 - Submission: SMTP over TLS (port 587)
 - Relay: SMTP with STARTTLS (port 25)
 - Retrieval: IMAP over TLS (port 993)

Dovecot



■ Installation

- `sudo yum install dovecot`

■ Configuration 1

- `sudo vim /etc/dovecot/dovecot.conf`

```
protocols = imap pop3 lmtp submission  
listen = *, ::
```

■ Configuration 2

- `sudo vim /etc/dovecot/conf.d/10-mail.conf`

```
mail_location = maildir:~/Maildir
```

■ Configuration 3

- `sudo vim /etc/dovecot/conf.d/10-auth.conf`

```
disable_plaintext_auth = no  
auth_mechanisms = plain login
```

■ Configuration 4

- `sudo vim /etc/dovecot/conf.d/10-master.conf`

```
unix_listener /var/spool/postfix/private/auth {  
    mode = 0666  
    user = postfix  
    group = postfix  
}
```

■ Start the server

- `systemctl enable dovecot`
- `systemctl start dovecot`

■ Firewall Configuration

- `sudo firewall-cmd --add-service=imap --permanent`
- `sudo firewall-cmd --add-service=pop3 --permanent`
- `sudo firewall-cmd -reload`